MIL-S-51482A(EA)
29 May 1986
SUPERSEDING
MIL-S-51482(EA)
16 June 1981

MILITARY SPECIFICATION

SIMULATOR, DETECTOR TICKETS, CHEMICAL AGENT: TRAINING, M256

This specification is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of training aid intended for training personnel in the use of Detector Kit, Chemical Agent: M256.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat-Sealable.
MIL-D-51083 - Detector Ticket, Anticholinesterase Agent.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 6910

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STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Test Procedures for Packaging Materials.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-810 - Environmental Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 5-77-2550 - Simulator, Detector Tickets, Chemical Agent: Training, M256

Special Packaging Instructions (SPI)

SPI P5-77-2550 - Simulator, Detector Tickets, Chemical Agent: Training, M256

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

MUNSELL BOOK OF COLOR, MATTE FINISH

(The Munsell Book of Color may be obtained from the Munsell Color Co., Inc. 2441 N. Calvert Street, Baltimore, MD 21218.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-77-2550 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the simulator shall conform to the specifications and drawings listed on TDPL 5-77-2550 and subsidiary drawings. The samplers contained in the simulator shall not be older than three months (from their date of manufacture).

3.2 Manufacture and assembly. The simulator shall be manufactured and assembled in accordance with Drawing 5-77-2550 and subsidiary drawings.

3.3 Ampoule leakage. The ampoules to be used in the simulator shall not leak when tested as specified in 4.4.4.1.

3.4 Potassium carbonate ampoule high temperature stability. The potassium carbonate ampoules to be used in the simulator shall withstand high temperature exposure without cracking or breaking when tested as specified in 4.4.4.2.

3.5 Packaged sampler leakage. The packaged samplers in the simulator shall not leak when tested as specified in 4.4.4.3.

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3.6 Cholinesterase activity. The impregnated and dried nerve spots, except those in T-401 samplers, shall contain 1.5 to 5.5 cholinesterase activity unit before and after storage at 90°C for 48 hours when tested as specified in 4.4.4.4.

3.7 Nerve spot color. The nerve spot shall have a uniform color and shall be no darker than Munsell Color Standard 10 YR 9/1.

3.8 Sampler performance. The sampler test spots shall meet the Munsell Color Standard equivalency requirements listed in table I when tested as specified in 4.4.4.5.

TABLE I. Sampler performance

Sampler	Munsell color standard equivalency requirement			
	Nerve spot	Blood spot	Blister spot	Lewisite tablet mark
T-400	2.5 B 9/2 or darker blue	WHITE	WHITE	10 YR 6/4 or darker red
T-401	2.5 YR 8/8 or lighter orange	WHITE	WHITE	10 YR 6/4 or darker red
T-402	2.5 B 9/2 or darker blue	WHITE	5 PB 8/6, 5 PB 7/8, or darker blue	10 YR 6/4 or darker red
T-403	2.5 B 9/2 or darker blue	WHITE	10 P 8/6, 10 P 7/8, or darker purple	10 YR 6/4 or darker red
T-404	2.5 B 9/2 or darker blue	7.5 PB 8/4, 5 PB 8/6, or darker blue	WHITE	10 YR 6/4 or darker red
T-404A	2.5 B 9/2 or darker blue	10 P 7/8, 10 P 6/10, or darker purple	WHITE	10 YR 6/4 or darker red

3.9 Workmanship. The simulator and all components and subassemblies thereof shall be free from defects and from contamination by foreign matter such as dirt, grease, oil, and dyes and chemicals contained in the ampoules.

3.10 Preproduction. Prior to the start of regular production, a preproduction sample quantity of simulators shall be produced in accordance with this specification for examination and test (see 4.3).

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3.10.1 High altitude. The simulator shall meet the sampler performance requirements of this specification after being subjected to high altitude pressure as specified in 4.3.3.1. There shall be no evidence of broken or leaking ampoules or disintegrated pellets.

3.10.2 Accelerated aging. The simulator shall meet the sampler performance requirements of this specification after being subjected to accelerated aging as specified in 4.3.3.2. There shall be no evidence of broken or leaking ampoules or disintegrated pellets.

3.10.3 Cold storage stability. the simulator shall meet the sampler performance requirements of this specification after being subjected to cold storage as specified in 4.3.3.3. There shall be no evidence of broken or leaking ampoules or disintegrated pellets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3).
(b) Quality conformance inspection (4.4).

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 100 simulators shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. The preproduction lot shall contain no more than one lot of each component, material, and reagent.

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4.3.2 Sampling and inspection procedure.

4.3.2.1 For examination. Sample preproduction simulators and components shall be examined in accordance with the classification of defects in 4.4.3.1 and MIL-STD-105. Examination after conditioning shall be as specified in Table II.

TABLE II. Preproduction inspection

Requirements	Sample size	Accept/reject criteria	Test paragraph
Ampoule leakage (3.3)	MIL-STD-105, prior to assembly in samplers	AQL 0.40 percent defective	4.4.4.1
Potassium carbonate ampoule high temperature stability (3.4)	MIL-STD-105, inspection level S-4, prior to assembly in samplers	AQL 4.0 percent defective	4.4.4.2
Packaged sampler leakage (3.5)	6 each T-400, T-401, T-402, T-403, T-404, and T-404A packaged samplers (36 total)	Acc 0 Rej 1	4.4.4.3
Cholinesterase activity (3.6)	4 each T-400, T-402, T-403, T-404, and T-404A packaged samplers (20 total)	Acc 1 Rej 2	4.4.4.4
High altitude (3.10.1), accelerated aging (3.10.2), and cold storage stability (3.10.3):	1 simulator for each condition		4.3.3.1, 4.3.3.2, 4.3.3.3
Examine samplers after conditioning for broken or leaking ampoules and disintegrated pellets	36 samplers for each condition	Acc 2 Rej 3 for each condition	
Test samplers after conditioning for performance (3.8)	4 each T-400, T-401, T-402, T-403; and 2 each T-404 and T-404A samplers for each condition (20 tests for each condition)	Acc 1 Rej 2 for each condition	4.4.4.5

4.3.2.2 For test. Sample preproduction simulators and components shall be tested as specified in table II.

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4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 High altitude. Subject the sample simulators to a pressure equivalent to 50,000 feet of altitute (3.44 inches of mercury) for no less than 1 hour in accordance with method 500.1, procedure I of MIL-STD-810.

4.3.3.2 Accelerated aging. Place the sample simulators in an oven at 90° ± 3°C for 48 ± 1 hour. Remove the simulators from the oven and allow them to reach ambient temperature.

4.3.3.3 Cold storage stability. Store the sample simulators at -35° ± 3°C for 48 ± 1 hour. Remove the simulators from cold storage and allow them to reach ambient temperature.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the simulators offered for acceptance at one time which have been produced by one manufacturer, at one plant, under the same manufacturing conditions, and from the same materials.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling shall be conducted in accordance with table III.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample kits and components shall be examined in accordance with MIL-STD-105 and the following classification of defects:

4.4.3.2 Classification of defects.

(a) Nerve spot color (see 3.7).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Color not uniform
102	Color incorrect

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TABLE III. Quality conformance inspection

Requirements	Sample size	Accept/reject criteria	Test paragraph
Ampoule leakage (3.3)	MIL-STD-105, prior to assembly in samplers	AQL 0.40 percent defective	4.4.4.1
Potassium carbonate ampoule high temperature stability (3.4)	MIL-STD-105, inspection level S-4, prior to assembly in samplers	AQL 4.0 percent defective	4.4.4.2
Packaged sampler leakage (3.5)	MIL-STD-105, inspection level S-4	AQL 2.5 percent defective	4.4.4.3
Cholinesterase activity (3.6)	MIL-STD-105, inspection level S-1. Exclude T-401 samplers.	AQL 2.5 percent defective	4.4.4.4
Sampler performance (3.8)	MIL-STD-105, inspection level S-1.	AQL 2.5 percent defective	4.4.4.5

(b) Sampler sub-assembly (Drawing 5-77-2553).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component missing, incorrect, or incorrectly assembled
102	Ampoules cracked, broken, or leaking
103	Printed instructions missing, incorrect, or illegible
104	Workmanship (see 3.9)

(c) Sampler assembly (Drawing 5-77-2554).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component missing, incorrect, or incorrectly assembled
102	Heater assembly broken or leaking
103	Workmanship (see 3.9)

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(d) Box assembly (Drawing 5-77-2562).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Component missing, incorrect, or incorrectly assembled
102		Box crushed or damaged
103		Workmanship (see 3.9)

(e) Bag (Drawing 5-77-2552).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Printed instructions missing, incorrect, or illegible
102		Workmanship (see 3.9)

(f) Instruction card (Drawing 5-77-2568).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Printed instructions missing, incorrect, or illegible
102		Card physically damaged
103		Workmanship (see 3.9)

(g) Simulator, detector tickets, chemical agent, training, M256 (Drawing 5-77-2550).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Component missing, incorrect, or incorrectly assembled
102		Packaging samplers ripped, torn, or punctured
103		Workmanship (see 3.9)

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(h) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Unit packing packing component missing, incorrect or incorrectly assembled
102	Unit packing or packing component damaged
103	Marking missing, incorrect, or illegible
104	Unitization incorrect (when required)
105	Workmanship (see 3.9)
106	Shelf life code marking missing, incorrect or illegible

4.4.3.2 For test. Sample simulators and components shall be tested as specified in table III.

4.4.4 Tests. Munsell color observations shall be made under north sky daylight as provided by a MacBeth Executive Daylight Lamp BBX-324 or equivalent. Tests shall be conducted as follows:

4.4.4.1 Ampoule leakage. Place the ampoules in a vacuum tank and weight with a suitable cover to ensure complete submersion when the tank is filled with water. Place sufficient water soluble dye in the water to impart visible color to a drop of solution. Reduce the pressure in the tank to 5 inches of mercury or less absolute, maintain at that pressure for at least 25 minutes, and then pressurize to ambient. After no less than 20 minutes at ambient pressure, remove the ampoule from the tank, rinse thoroughly in clean water, and spread on a drying screen. Colored water in an ampoule shall be considered evidence of leakage.

4.4.4.2 Potassium carbonate ampoule high temperature stability. Place the ampoules in an oven and subject them to a temperature of 150° ± 5°C for 90 ± 5 minutes. After the test, examine the ampoules for cracks or breaks. Cracked or broken ampoules shall be considered defects. Ampoules subjected to this test shall not be used in the manufacture of simulators.

4.4.4.3 Packaged sampler leakage. Conduct the leak test in accordance with method 5009.1, vacuum chamber technique of Fed. Test Method Std. No. 101.

4.4.4.4 Cholinesterase activity. The samplers selected for testing shall be packaged in accordance with Drawing 5-77-2551. Half of these samplers shall be stored at 90° ± 2°C for 48 hours before testing. After storage, determine the cholinesterase activity of the stored and unstored nerve gas test spots in accordance with the cholinesterase activity test method specified in MIL-D-51083 except that manually operated buret or pipet and a laboratory pH meter may be substituted for the pH stat titrator.

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4.4.4.5 Sampler performance. Protective strip should still cover spots. Pull and discard tab numbered "1" to expose tablet. Rub top half of white paper tab numbered "2" on tablet. Hold sampler with arrow up. Using heater pads, crush ampoules in three center pockets numbered "3". Rotate sampler so arrow is down. Force liquid to each spot with right hand while pressing protective strip with left hand to ensure wetting of covered spots. With left thumb over center of protective strip, hold sampler flat and swing heater (green ampoules) away from blister spot. Being sure NOT TO use pads, crush one heater ampoule numbered "4" and swing heater over spot. Vent heater vapor away from operator's body. Leave heater over spot for 2 minutes, then swing heater and protective strip away from spots. Expose spots to air (shielding from direct sun) for 10 minutes. Crush second green ampoule numbered "4" and swing heater back over test spot. Leave in place 1 minute. Swing heater away from spot. Hold sampler with arrow down. Crush two outside ampoules numbered "5" with pads. Wait 2 minutes and record the Munsell color or colors which most nearly match the colored portion of the spots and rubbing tab.

5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in Special Packaging Instruction (SPI) P5-77-2550.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation required (see 5.1).
(c) Level of packaging required (see 5.1).
(d) If unitization is required (see 5.1).
(e) Shelf life code markings (see 5.1).

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6910-A066

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-51482A(EA)	SIMULATOR, DETECTOR TICKETS, CHEMICAL AGENT: TRAINING, M256

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

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